Full-Throttle Speed Loss in 2026 Formula One Qualifying: A Measurement from Public Telemetry

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October 2026

Abstract

The 2026 Formula One technical regulations nearly triple the power of the kinetic motor generator unit (MGU-K), to 350 kW, and reduce that power with speed above 290 km/h. A visible consequence is *clipping*, in which cars lose speed on a straight while the driver remains at full throttle. We measure this effect in every 2026 qualifying session held up to the Singapore Grand Prix, using public FastF1 telemetry cross-checked against OpenF1 and cleaned using documented sentinel values. Across 41 reportable straights in 16 sessions (842 driver-straights), 86% of driver-straights reach their peak speed before the end of the full-throttle segment. Differences between power-unit manufacturers are small: the five manufacturer medians span 2.1 km/h, comparable to the median gap between teammates (2.0 km/h). On 39 straights matched between 2025 and 2026 at 15 circuits, all 33 straights that show clipping have an earlier peak and a larger speed loss in 2026, while the remaining 6 show no clipping in either year.

1 Introduction

The 2026 Formula One regulations shifted a much larger share of the car's power to the electrical system. The MGU-K may now deliver 350 kW rather than 120 kW, and the heat motor generator unit (MGU-H), which previously recovered energy from the turbocharger, has been removed (Fédération Internationale de l'Automobile, 2026; Medland, 2026). With more electrical power to supply and fewer ways to recover energy, the cars depend heavily on harvesting kinetic energy, and the regulations limit both the energy recovered per lap and the electrical power available at high speed. From pre-season testing onwards, drivers and commentators reported cars that stopped accelerating, or slowed, before the braking point while the driver remained at full throttle (Noble, 2026b,a; F1Technical, 2026a).

Speed lost at full throttle is lap time that cannot be recovered on that straight, and the effect quickly became a sporting and safety concern. After the first three rounds, the FIA, teams and power-unit (PU) manufacturers agreed changes to the energy rules for qualifying, which applied from the Miami Grand Prix (FIA, 2026a). Public discussion of the effect has so far relied on individual laps, single corners and broadcast speed traces (Noble, 2026a; F1Technical, 2026a).

Energy deployment under a per-lap budget has been studied extensively as an optimal-control problem for the 2014–2025 hybrid power unit. These studies show that the lap-time-optimal strategy distributes electrical energy unevenly around the lap, concentrating it where it saves the most time (Limebeer et al., 2014; Salazar et al., 2017; Ebbesen et al., 2018), but they rely on detailed vehicle models and team data. Analyses based on public data are fewer. Bopaiah and Samuel (2020) reconstructed energy-recovery use from on-board broadcast graphics, and in later work compared drivers, teams sharing a PU and the whole field within a single qualifying session, using positive kinetic energy as a proxy for deployment (Bopaiah and Samuel, 2021). Statistical studies of Formula One performance have generally worked from race results rather than telemetry (Bell et al., 2016; van Kesteren and Bergkamp, 2023).

This paper measures full-throttle speed loss across the 2026 season using public timing and telemetry data. As in Bopaiah and Samuel (2021), reconstructed lap distance is validated against the official track length. The analysis extends that approach to a full season, reports continuous measurements in physical units, records the reason for every excluded lap and segment, and adds a matched comparison with 2025. It addresses three questions:

1. Where on a straight do 2026 cars reach their peak speed, and how much speed do they lose after it?
2. How large are the differences between PU manufacturers compared with the differences between teammates?
3. How do the same straights compare between 2025 and 2026?

Section 4 addresses the first two questions using 2026 data, and Section 5 the third.

2 Background

2.1 From the 2014 hybrid power unit to 2026

From 2014 to 2025, the Formula One power unit combined a 1.6-litre turbocharged engine with two motor-generators. The MGU-K, on the drivetrain, recovered energy under braking and returned it as propulsion, while the MGU-H recovered energy from the exhaust through the turbocharger. The MGU-K was limited to 120 kW and could recover at most 2 MJ per lap, but energy flow through the MGU-H was unrestricted, allowing up to 4 MJ per lap to be deployed from the energy store (Bopaiah and Samuel, 2020, 2021; Limebeer et al., 2014).

The 2026 regulations remove the MGU-H. Only the engine and the kinetic energy recovery system (ERS-K) may propel the car or recover energy (Article C5.2.1), and the electrical power of the ERS-K rises to 350 kW (C5.2.7) (Fédération Internationale de l'Automobile, 2026).[1] The engine now supplies roughly half of the total power (Medland, 2026), and the ERS-K may recover up to 8.5 MJ per lap (C5.2.10), more than four times the previous MGU-K limit. Since this energy can come only from the MGU-K, it must be harvested from the car's own motion: under braking, on partial throttle, or by running the MGU-K as a generator against the engine. The 2026 cars are also lighter and smaller. DRS has been replaced as the overtaking aid by Overtake, which provides additional electrical power at high speed (C5.2.8 ii), and the front and rear wings switch between a "Corner Mode" and a lower-drag "Straight Mode" (C3.10, C3.11) (Fédération Internationale de l'Automobile, 2026; Medland, 2026). Table 1 summarises the changes relevant to this study.

Table 1: Power-unit and car rules relevant to full-throttle speed loss, 2025 and 2026.

	2014–2025	2026
MGU-K maximum power	120 kW	350 kW, reduced above 290 km/h
MGU-K energy recovered per lap	2 MJ	8.5 MJ, lower at some events
MGU-H	Fitted; energy unrestricted	Removed
Drag reduction	DRS (rear wing)	Straight Mode (both wings)
Overtaking aid	DRS	Overtake (extra MGU-K power)

Sources: Bopaiah and Samuel (2020, 2021); Fédération Internationale de l'Automobile (2026); Medland (2026).

[1] Article numbers and values refer to Issue 20 of Section C (5 August 2026). Compared with Issue 12 (June 2025), every value used here is unchanged. Two modes have been renamed (Override Mode is now Overtake, and Straight-Line Mode is now Straight Mode), and the event-level reductions of the recovery limit in C5.2.10 now reach 7 MJ, and 4 MJ in qualifying.

2.2 Power reduction at high speed

Above a certain speed, the permitted MGU-K power is reduced. Article C5.2.8 i limits the MGU-K power used to propel the car to $P = 1800 - 5v$ kW for $v < 340$ km/h, to $6900 - 20v$ kW for $340 \leq v < 345$ km/h, and to zero at 345 km/h and above (Fédération Internationale de l'Automobile, 2026). Combined with the 350 kW cap, this gives full electrical power up to

$$v_{\text{onset}} = \frac{1800 - 350}{5} = 290 \text{ km/h}, \tag{1}$$

above which the permitted power falls by 5 kW per km/h, reaching 100 kW at 340 km/h and zero at 345 km/h (Figure 1). Overtake, whose use is governed by the Sporting Regulations, shifts the curve to higher speeds (C5.2.8 ii). Many straights on the 2026 calendar exceed the onset speed. At Suzuka, for example, peak speeds on the three measured straights ranged from 281 to 338 km/h; the T12→T13 straight peaks close to the onset and the other two well above it.

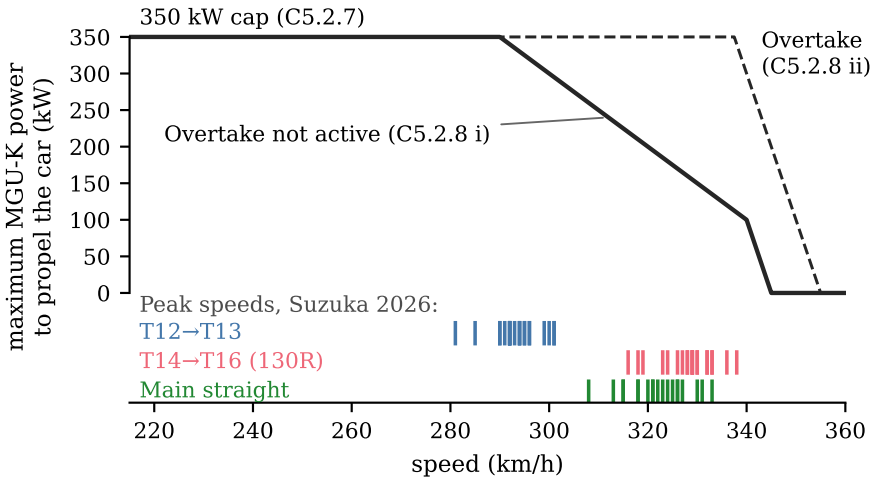


Figure 1: Maximum MGU-K propulsion power against speed under Article C5.2.8 (solid) and with Overtake active (dashed). Ticks mark the peak speeds on the three Suzuka straights measured in 2026.

2.3 Clipping and superclipping

On a straight, a car accelerates as long as its propulsive power exceeds the power needed to overcome aerodynamic drag and rolling resistance. Because drag force grows with the square of speed, the power it absorbs grows with the cube (Milliken and Milliken, 1995). When propulsive power falls below this requirement, the car slows even with the throttle fully open. In 2026 this can happen in three ways: the speed-dependent limit can reduce the electrical power; the car can run out of the energy it is permitted, or has chosen, to deploy on that straight; or the MGU-K can be switched into generator mode, taking power from the engine to recharge the battery. The second is commonly called *clipping*, a term that predates 2026, and the third *superclipping* (Mitchell-Malm and Straw, 2026; Noble, 2026b; RACER, 2026). Superclipping is typically used towards the end of straights, where recovering energy costs the least lap time, and its peak power is capped by the regulations (Noble, 2026b; FIA, 2026a).

These mechanisms cannot be distinguished in public telemetry. The car-data stream carries six channels (Speed, RPM, gear, Throttle, Brake and DRS) at roughly 4 Hz (Schaefer, 2026; OpenF1, 2026), and none of them describes the energy system: there is no MGU-K power or torque, no state of charge, and no indication of whether the MGU-K is deploying or harvesting. In this paper, *clipping* therefore refers to the observed outcome, a fall in speed within a full-throttle segment, regardless of its cause.

2.4 In-season changes and the Miami split

Driver criticism of qualifying in the opening rounds focused on part-throttle running and long superclipping phases. In one widely reported case, a car lost up to 15 km/h over much of a straight compared with the same driver's previous run (Noble, 2026a). The FIA first lowered the maximum energy that could be recharged in qualifying from 9 to 8 MJ for the Japanese Grand Prix (F1Technical, 2026b). On 20 April, the FIA, teams, PU manufacturers and Formula One Management agreed a wider package that took effect from the Miami Grand Prix (FIA, 2026a). For qualifying, the maximum recharge fell from 8 to 7 MJ, with the stated aim of limiting superclipping to roughly two to four seconds per lap, and peak superclipping power rose from 250 to 350 kW. Lower energy limits could also be applied at 12 events instead of 8.

Both changes bear directly on the quantity measured here, since less energy to recover should shorten superclipping phases and a higher harvesting power makes each phase shorter but sharper. The season is therefore divided into a pre-Miami period (Australia, China and Japan; 3 sessions) and a post-Miami period (14 sessions). Because the two periods share no circuit and also differ in data quality (Section 3.3), differences between them are reported but not attributed to the rule changes (Section 7).

3 Data and measurement

3.1 Sources and sessions

Telemetry is taken from FastF1 version 3.8.3 (Schaefer, 2026), an open-source Python package that provides the official Formula One live-timing feed, including car data (Speed, RPM, gear, Throttle, Brake) and position data (X/Y), and which has been used in recent peer-reviewed work (Cappello and Hoegh, 2026). Brake values are cross-checked against the `car_data` endpoint of OpenF1 (OpenF1, 2026), which was available for every session used.

The 2026 calendar was announced with 24 rounds. The Bahrain and Saudi Arabian Grands Prix, scheduled for April, were cancelled in March because of the conflict in the Middle East, leaving 22 rounds (Sky Sports, 2026). The Bahrain Grand Prix was later restaged at Sepang, Malaysia, in October, bringing the season to 23 rounds (Formula 1, 2026). All seventeen qualifying sessions held up to the Singapore Grand Prix (10 October 2026) are included. FastF1's circuit information contains no corner positions for the new Madrid circuit or for Sepang, which last hosted Formula One in 2017, so straights at these circuits cannot be labelled by the corner at which they end. Sepang therefore contributes only its main straight, and Madrid contributes no reportable straight. For 2025, we use the 15 qualifying sessions held at circuits that also appear on the 2026 calendar, matched by OpenF1 `circuit_key`.

3.2 Validation of distance

Following Bopaiah and Samuel (2021), we compared each driver's fastest-lap distance at Suzuka 2026 with the official track length of 5.807 km. Distance integrated from X/Y position fell within -0.83 to -0.43% of the official length (mean absolute error 0.58%), while distance integrated from car-data speed had a mean absolute error of 1.48% and a maximum of 7.37%. Bopaiah and Samuel (2021) report a mean error below 0.6% and a maximum below 1.8% for their reconstructed data. The larger car-data errors are caused by frozen samples, described below.

3.3 Frozen samples

Car data occasionally freeze, with Speed, RPM and gear held constant while the car keeps moving. On the 2026 Suzuka pole lap, for example, car data froze at 189 km/h for 25.8 s. The same freezes appear in OpenF1, indicating that they originate upstream of FastF1. They are unevenly distributed across sessions (Figure 2). In 2025, the share of frozen samples on push laps

did not exceed 0.06% at any paired circuit, whereas in 2026 it reached 10.6%, with Shanghai, Suzuka and Monza all above 1%. The mean freeze share was 7.0% before Miami and 0.2% from Miami onwards. Segments that overlap a freeze are excluded. Freezes are identified only from values documented by FastF1, without thresholds tuned to the data; the full rules are given in Appendix A.

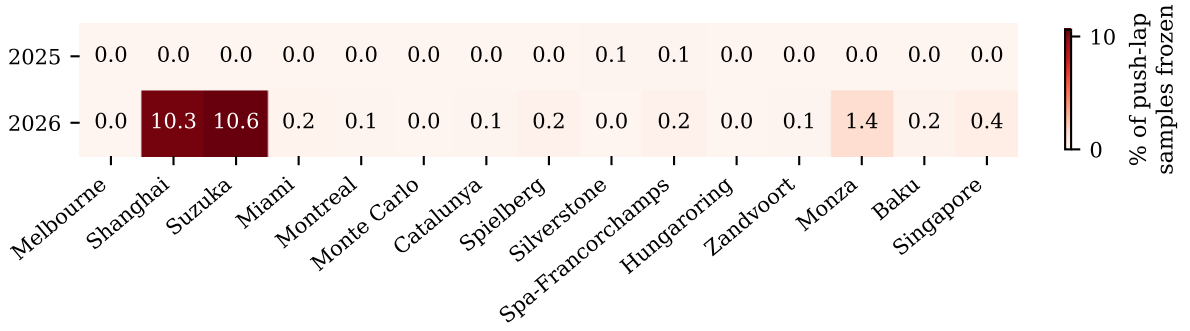


Figure 2: Share of push-lap car-data samples that are frozen, per paired circuit (2026 calendar order) and year.

3.4 Push laps and full-throttle segments

Push laps are laps that do not involve the pit lane, were not deleted by the stewards, and are within 107% of the session's fastest lap time. The 107% threshold is FastF1's default for identifying representative laps (Schaefer, 2026) and mirrors the qualifying cut-off in the Sporting Regulations (FIA, 2026b); it removes out-laps, cool-down laps and aborted laps, which could otherwise be mistaken for clipping. We also discard laps whose speed is zero throughout, whose X/Y position is $(0, 0)$ on any sample, or whose position does not change.

A *full-throttle segment* is a run of consecutive samples, at least 500 m long, with the throttle at 98% or more and the brake off or unknown. FastF1 records Throttle as pedal position from 0 to 100% (Schaefer, 2026), and a threshold slightly below 100 prevents a straight from being split by readings marginally below full pedal. For the same reason, a single sample below the threshold does not end a segment: at about 4 Hz, one sample covers roughly 20 m at 300 km/h, which is too short to be a deliberate lift. The 500 m minimum, about 25 samples at that speed, retains genuine straights and excludes short bursts of throttle between corners, where the car is still accelerating and the position of the peak carries little information.

To check that these choices do not drive the results, the analysis was repeated for throttle thresholds of 95, 98 and 100% and minimum lengths of 300, 500 and 800 m (Appendix B). In all nine combinations, most driver-straights peak before the segment end and the median paired 2025–2026 difference is positive. The only exception to the straight-by-straight count is Silverstone T16, a segment of about 310 m that qualifies only at the 300 m minimum and whose median loss is below 1 km/h in both years. The throttle threshold has little effect, whereas the minimum length changes the magnitudes: longer minimums retain only long straights, which peak earlier and lose more speed, reflecting the confounding with straight length discussed in Section 4.1.

Each segment is labelled by the corner at which it ends, matched by X/Y position to FastF1's corner list (Figure 3), and the main straight is stitched across the timing line. For each driver and straight we keep one lap: the fastest lap with a clean segment on that straight, taken from the latest qualifying part in which one exists. A straight is *reportable* in a session if at least 10 drivers have a clean segment on it. This gives 41 reportable straights in 16 sessions (842 driver-straights), 8 of them before Miami and 33 from Miami onwards.

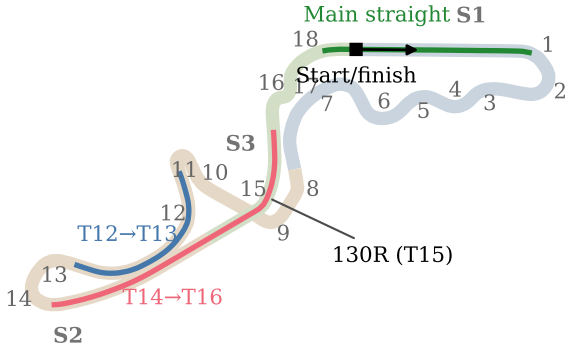


Figure 3: Suzuka, 2026: FIA sectors, corner numbers, start/finish with direction of travel, and the three measured straights.

3.5 Segment metrics

For each selected segment, with distance d from its start and length L, we record:

- `peak_speed`: the maximum speed in the segment, taken at the last sample if the maximum is reached more than once;
- `vmax_position` $= d_{\text{peak}}/L$: where the peak occurs, from 0 (segment start) to 1 (segment end), so that a value below 1 means the car reached its top speed before the driver lifted or braked;
- `clip_delta` = `peak_speed` − `end_speed`: the *speed lost after the peak* (km/h), up to the last full-throttle sample;
- `decel_rate` $= 100 \cdot$ `clip_delta`$/(L - d_{\text{peak}})$: the speed lost per 100 m after the peak.

Because speed is reported in whole km/h, a car holding a constant, drag-limited top speed can register a peak slightly before the segment end with no real loss. We therefore also report the share of drivers with `clip_delta` ≥ 2 km/h.

Results are summarised as counts, medians and interquartile ranges (IQR). We do not report significance tests: with one lap per driver and several straights per circuit, the observations are not independent, and p-values would overstate the strength of the evidence.

4 The 2026 season

This section uses all 842 driver-straights. Section 4.1 uses raw values. For the comparison between PU manufacturers in Section 4.2, each straight's mean over all its clean drivers is first subtracted from each driver's value. This centring removes effects common to all cars on a straight, such as its length, gradient and the energy limit set for that event, leaving differences between cars and drivers. Centred values are then pooled by PU.

4.1 Most 2026 cars peak before the end of the straight

In 2026 qualifying, most cars reach their peak speed before the end of the full-throttle segment and then lose speed while still at full throttle. Of the 842 driver-straights, 86% peak before the segment end, with a median peak position of 0.76, about three-quarters of the way along the straight (Figure 4). Commentators have described 2026 cars as peaking earlier on a straight than their predecessors (The Race, 2026); Section 5.2 tests this against 2025 data.

The relationship between speed loss and peak speed is not monotonic (Figure 5). The median loss is higher for the 106 driver-straights that peak below the 290 km/h onset (13.0 km/h) than for the 736 that peak at or above it (11.0 km/h). Since the speed-dependent limit does not

apply below the onset, these losses have another source. Four of the five straights whose median peak is below the onset contain at least one corner taken flat out (Shanghai T8, Monaco T10, Spa-Francorchamps T12 and Baku T15), where drag and cornering load reduce speed whatever power is available; the fifth, the Montreal main straight, does not. Above the onset, speed loss increases with peak speed (Spearman $\rho = 0.36$; Spearman, 1904), which is consistent with the speed-dependent limit. Peak speed is, however, confounded with straight length, since longer straights allow both higher peak speeds and more distance over which to lose speed, and circuit is confounded with both.

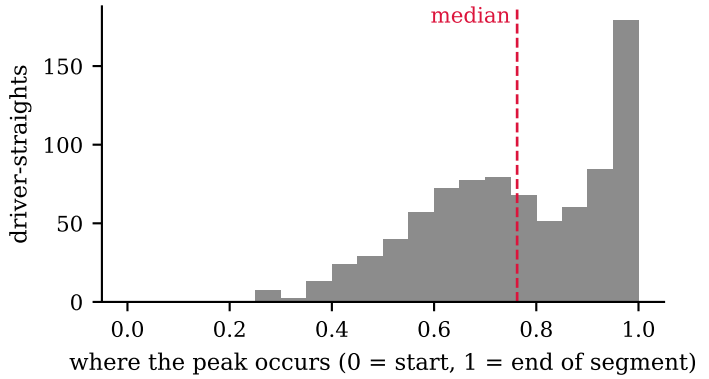


Figure 4: Where on the straight the peak speed occurs, all 842 driver-straights on 41 reportable straights. Dashed line: median (0.76).

4.2 PU differences are about as large as teammate differences

Teammates share a car and a power unit, so the difference between them reflects driving and lap-to-lap variation rather than the car, and provides a useful reference for differences between manufacturers. Results-based studies use the same logic to separate driver from constructor effects (Bell et al., 2016; van Kesteren and Bergkamp, 2023). After centring, the spread between PU manufacturers is similar to the gap between teammates (Figure 6). The PU medians of centred `clip_delta` range from −0.9 km/h (Mercedes) to +1.1 km/h (Honda), a span of 2.1 km/h, while teammates on the same straight differ by a median of 2.0 km/h (IQR 1.0–4.0, 408 pairs). Team medians within a single PU span up to 3.8 km/h, and the PU medians of centred `decel_rate` range from −0.49 to 0.29 km/h per 100 m.

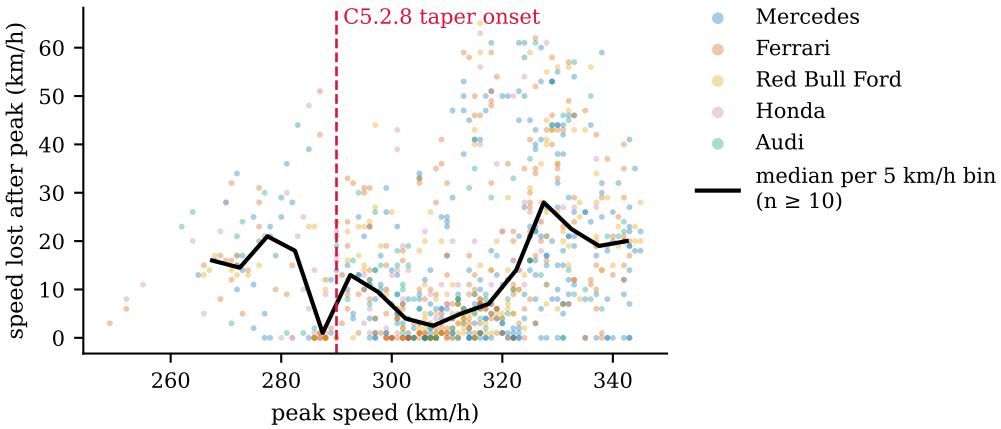


Figure 5: Speed lost after the peak against peak speed, one point per driver-straight, coloured by PU. Black line: median per 5 km/h bin with at least 10 points (3 sparser bins with 8 points in total omitted). Dashed: 290 km/h onset.

Splitting the straights by speed regime gives a similar picture. Here each straight is assigned as a whole according to its median peak speed, so the counts differ slightly from the per-driver split in Section 4.1. On the 5 straights with a median peak below 290 km/h (88 driver-straights), PU medians range from −0.4 to 2.9 km/h; on the 36 straights at or above 290 km/h (754 driver-straights), they range from −0.9 to 2.4 km/h. With one or two cars per team and one lap per driver, the sample is too small to rank the manufacturers.

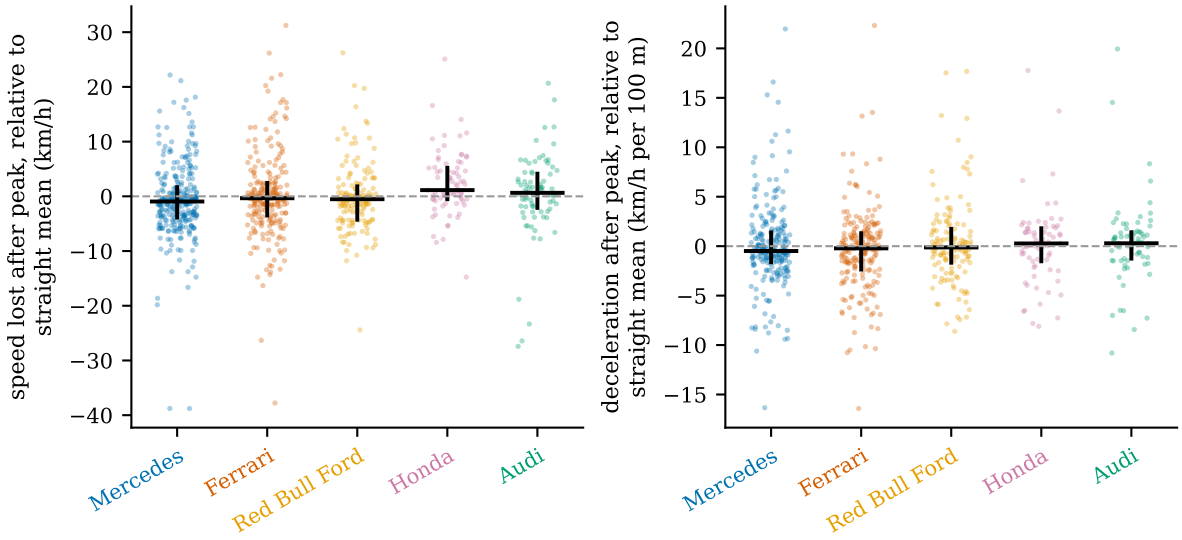


Figure 6: Centred speed lost after the peak (left) and deceleration after the peak (right) by PU. Dots: driver-straights; bar: median; line: IQR; 0: straight mean.

5 Comparison with 2025

5.1 Design

A straight is *matched* if the same corner label exists at the same circuit in both years, with at least 10 clean drivers in each, which yields 39 matched straights at 15 circuits. A matched straight *shows no clipping* if its median `clip_delta` is zero in both years, and is counted as a straight *with clipping* otherwise.

For each matched straight and year, we take the median over drivers (one lap each), compute the 2026 − 2025 difference per straight, and report the median and IQR of these differences. Pairing each straight with itself holds the circuit, the corner and the length of the straight constant. All 2026 cars are pooled, given the small differences between PUs found in Section 4.2.

We ran two robustness checks. First, because a changed braking point or a mislabelled corner would move the end of a segment, we flag any straight whose median segment end point moved by more than 50 m between years, measured from position data. Second, we repeat the comparison without any circuit at which FastF1 weather data record rainfall at any point in qualifying in either year. Rain lowers grip and speed and changes lines, tyre choice and where a driver can stay flat out (Pirelli, 2020); from Miami onwards, wet sessions also run with reduced maximum ERS deployment (FIA, 2026a), and the Sporting Regulations treat wet qualifying differently, for instance by suspending the 107% rule (FIA, 2026b).

5.2 Every straight that clips does so earlier and harder in 2026

On all 33 matched straights with clipping, 2026 cars peak earlier and lose more speed after the peak than 2025 cars did on the same straight. The other 6 matched straights show no clipping in either year.

Speed signature. Figure 7 pools the matched straights into a single speed profile per year, with each driver's segment rescaled to a common length and its speed divided by its own peak, so that straights of different lengths and speeds can be overlaid. In 2025, the median curve rises all the way to the end of the segment, and the median peak position over driver-straights is 1.00 (0.99 in Table 2, which uses the median of the per-straight medians). In 2026, the curve rises faster, peaks at a median of 0.77 of the segment length and falls before the braking point. It lies above the 2025 curve from the start of the segment to 0.84 of its length, with a gap of 0.025 of peak speed at the start, and then crosses below it, ending at 96.4% of peak speed compared with 100.0% in 2025.

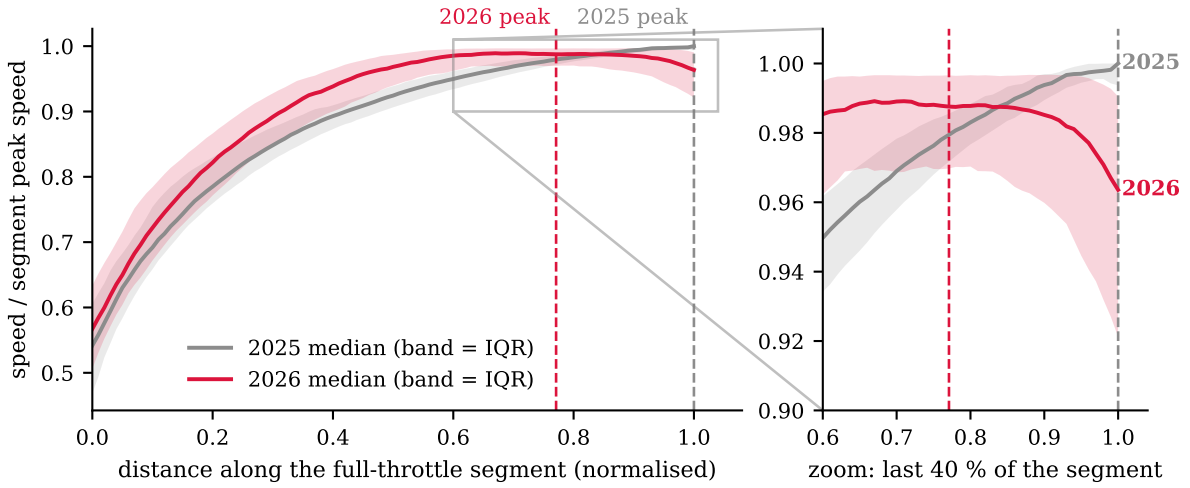


Figure 7: Speed signature of the 39 matched straights: speed relative to the segment peak against normalised distance. Line: median; band: IQR across driver-straights (762 in 2025, 801 in 2026). Dashed: median peak position. Right: zoom on the last 40% of the segment.

Straight by straight. The same pattern holds on each straight individually (Figure 8), where each row shows one matched straight with the 2025 and 2026 values joined, following recommendations for presenting paired data (Weissgerber et al., 2015). On every clipped straight, the 2026 point lies to the left in the peak-position panel and to the right in the speed-loss panel. Across the 33 straights, the peak moved towards the start by a median of 0.20 of the straight length (range 0.01 to 0.59), and the speed lost after the peak increased by a median of 10.5 km/h (range 1.0 to 50.5). Some changes are small, notably at Monza T4 (−0.01, +2.0 km/h) and on the Baku main straight (−0.01, +1.0 km/h), but each straight counts equally. Segments that include corners taken flat out lose some speed in 2025 as well, through drag and cornering load, which the pairing accounts for by comparing each straight with itself. Table 2 summarises the paired differences across all 39 straights.

Robustness. Neither check in Section 5.1 changes the result. Median segment end points moved by 0–32 m between years (median 12 m), so no straight was flagged. Rain fell during qualifying at Baku in 2025 and at Montreal and Zandvoort in 2026. Excluding these circuits leaves 31 matched straights; on all 26 with clipping, 2026 cars still peak earlier and lose more speed, the other 5 show no clipping in either year, and the median `clip_delta` difference remains +7.50 km/h.

Suzuka. Figure 9 illustrates the pattern on a track map, colouring each straight by the speed lost since its peak on the fastest freeze-free push lap of each year. The largest loss after a peak is 10 km/h in 2025 (VER, lap 16) and 50 km/h in 2026 (PIA, lap 16).

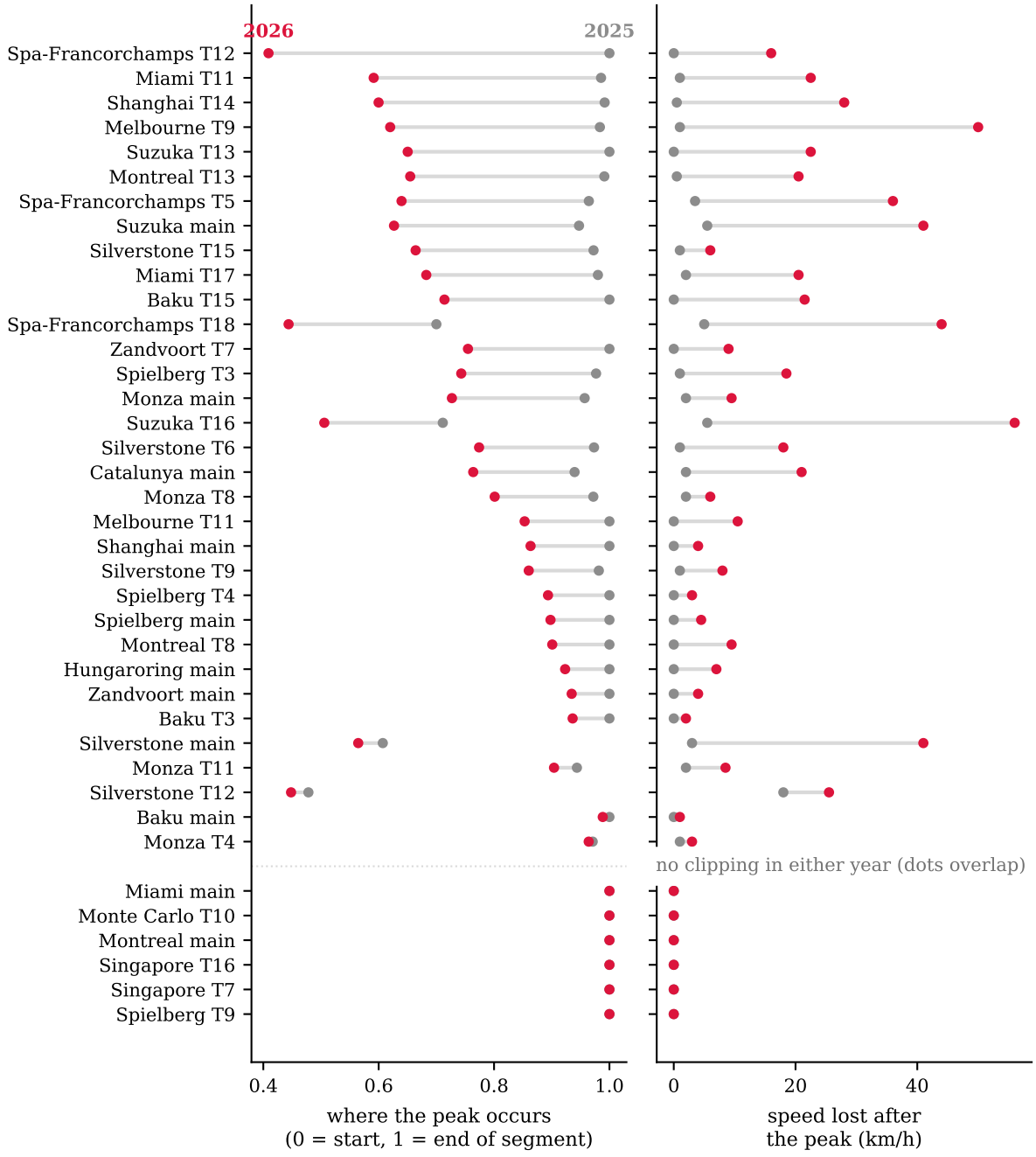


Figure 8: Each matched straight in 2025 (grey) and 2026 (accent). Each dot is the median across drivers, one lap each (13–20 drivers in 2025, 11–22 in 2026). Straights with clipping (33) are sorted by how far the peak moved; the 6 without clipping are at the bottom, where the dots overlap.

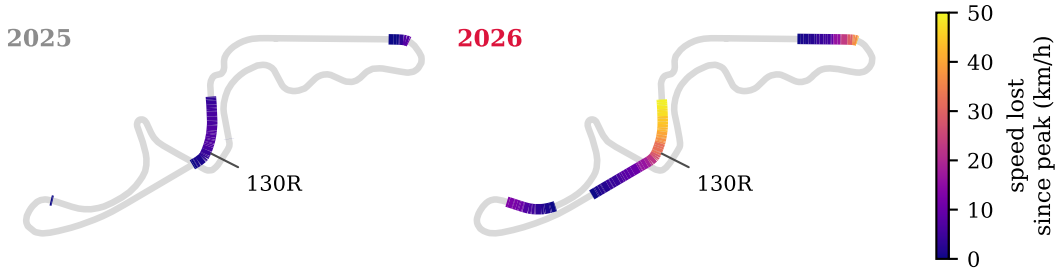


Figure 9: Suzuka, fastest freeze-free push lap per year, on one colour scale: speed lost since the segment's peak. Grey: outside full-throttle segments or before the peak.

Table 2: Paired difference 2026 − 2025 across 39 matched straights: median of the per-straight medians in each year, median and IQR of the per-straight difference, and share of straights where 2026 is higher. "Peak before end" and "loss ≥ 2 km/h" are shares of drivers on a straight.

Measure	Median 2025	Median 2026	Median diff. (IQR)	2026 > 2025
`vmax_position`	0.99	0.77	−0.15 (−0.29 to −0.04)	0%
`clip_delta` (km/h)	0.50	9.50	+7.50 (+3.50 to +20.75)	85%
Peak before end	0.50	1.00	+0.31 (+0.10 to +0.51)	87%
Loss ≥ 2 km/h	0.25	1.00	+0.40 (+0.16 to +0.68)	87%

Note. "Peak before end" rises on one more straight (87%) than median `clip_delta` does (85%). Because speed is reported in whole km/h, a driver at a drag-limited top speed can register a peak before the end with no measurable loss. At the Montreal main straight, for example, the share of drivers peaking before the end rose from 0.05 to 0.36 while the median `clip_delta` was 0 in both years.

6 Discussion

Full-throttle speed loss is common in 2026 qualifying, much larger than on the same straights in 2025, and not concentrated in any one manufacturer. The spread between PU manufacturers is no larger than the gap between teammates, which suggests that the behaviour is a property of the 2026 cars as a group, although with one or two cars per team smaller PU differences cannot be ruled out. Above the 290 km/h onset, speed loss tends to increase with peak speed, but this relationship is confounded with straight length.

The shape of the 2026 speed profile is consistent with what optimal-control studies predict for an energy-limited hybrid. When the energy available per lap is capped, the time-optimal strategy spends it where it saves the most time and accepts less electrical support elsewhere (Salazar et al., 2017; Ebbesen et al., 2018). A faster rise early in the straight followed by a fall before braking would be expected if electrical power is concentrated on acceleration and then withdrawn, or reversed into recovery, near the end of the straight, where speed and drag are highest. The present data are consistent with this explanation but cannot confirm it.

Bopaiah and Samuel (2021) studied a single qualifying session in detail. This study instead measures one quantity across a full season, with a matched 2025 comparison, using cleaning rules that were developed on one event (Suzuka) and then applied unchanged to the others. A detailed single-session analysis can show how a car uses its energy on a given lap, whereas a season-wide measurement shows how general a pattern is.

Separating the contributions of the speed-dependent power limit, the energy limits and team or driver choices is not possible with public telemetry. It would require the energy store's state of charge, the MGU-K's deployment and harvesting power along the lap, the energy recovered per lap relative to the event's limit, and the use of Overtake, none of which is available in the public feed. With these channels, taper-limited, energy-limited and harvest-driven speed loss could be separated straight by straight, and a model adjusting for straight length could replace the comparisons of medians used here.

7 Limitations

- **Statistical inference.** Results are reported as counts, medians and ranges rather than estimated effects. Interval estimates resampled by circuit, and a model adjusting for straight length, would be natural extensions.
- **Confounding.** Peak speed is confounded with straight length and circuit (Section 4.1). Energy limits are also set event by event, so part of the variation between circuits may reflect the limit in force rather than the circuit itself (FIA, 2026a).
- **The Miami split.** No circuit appears in both periods, and pre-Miami data quality was worse (mean freeze share 7.0% against 0.2%). Median `clip_delta` was 26.0 km/h on the

8 pre-Miami straights and 9.0 km/h on the 33 post-Miami straights. The direction is consistent with the lower qualifying recharge limit introduced at Miami, but the difference cannot be separated from circuit or data quality.

- **Whole-package comparison.** The 2025–2026 difference reflects every change between the two cars, including the aerodynamic modes and the lower mass, not the power-unit rules alone.
- **No energy-system channels.** Public telemetry contains no state of charge, deployment or harvesting power, so a given speed loss cannot be attributed to the speed-dependent limit, the energy limits, superclipping or driver choice.
- **Sample.** There are one or two cars per team and one lap per driver per straight. Madrid and Sepang have no corner-labelled straights.
- **Measurement.** Speed is reported in whole km/h, and the reported magnitudes depend on the minimum segment length (Appendix B). Freezes remove more data in some sessions (up to 10.6% at Suzuka 2026). The `JumpFlag` check (Appendix A) was run for Suzuka only, where 1 of 62 selected segments was flagged and excluding it changed the mean `clip_delta` per straight by at most 0.4 km/h. Its bound is derived from 2026 values and is not interpreted for 2025.

8 Conclusion

Public telemetry, cleaned using documented sentinel values, is sufficient to measure full-throttle speed loss across a Formula One season. In 2026 qualifying, most cars lose speed before the end of a straight, the effect is not specific to one manufacturer, and on every matched straight where it occurs it comes earlier and costs more speed than in 2025. Attributing the loss to the speed-dependent power limit, the energy limits or superclipping would require energy-system data that teams record but the public feed does not carry.

Data and code availability

The pipeline (`flatout.py`) and notebooks are available at

`https://github.com/see7ien/Flat-Out`

All numbers in this paper are taken from the notebook outputs.

A Cleaning rules

Freezes and unknown brake states are identified from documented sentinel values only, without thresholds tuned to the data. The one rule with a numerical bound, (c), flags segments but does not remove them.

(a) A run of unchanged Speed, RPM and gear that contains `Throttle` = 104 is treated as a freeze, together with a one-sample buffer on each side. FastF1 documents 104 as an error or unavailable value, usually seen when the car is stationary (Schaefer, 2026), and its maintainer notes that it should not occur in lap telemetry (Schaefer, 2025); in these data it also occurs while cars are running. Segments that overlap a freeze are excluded.

(b) OpenF1 `brake` = 104 indicates an unknown brake state. It does not end a full-throttle segment and is trimmed from the segment's end. OpenF1 `brake` = 100 is treated as genuine braking.

(c) A single-step speed increase at least 5 km/h larger than 800 kW could produce for a 700 kg car is flagged (`JumpFlag`) but not removed. Results are reported with and without flagged segments, following the logic of a multiverse analysis (Steegen et al., 2016).

A Speed/RPM ratio check (Segers, 2014, sec. 6.4) was also tested but did not separate stale from updating speed, so it is used only as a diagnostic. Figure A1 shows where the flagged samples fall on the Suzuka lap in 2026.

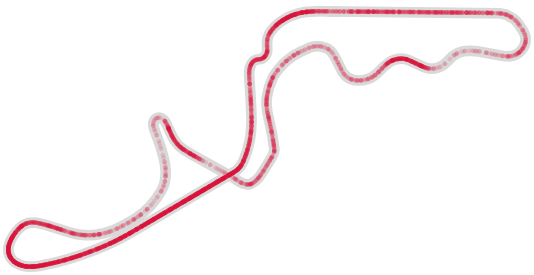


Figure A1: Locations of 3,940 flagged car-data samples (including one-sample buffers) on 75 of 108 push laps, Suzuka 2026.

B Sensitivity to the segment definition

Segment extraction and clean-lap selection were rerun with the same code for every combination of throttle threshold (95, 98 and 100%) and minimum segment length (300, 500 and 800 m), on all 2026 sessions and all paired 2025 sessions, with the reporting rule unchanged (at least 10 clean drivers per straight). Table B1 gives the headline numbers for each combination; the row in bold is the definition used in the paper.

Table B1: Headline numbers under nine segment definitions. Season columns use all reportable 2026 straights; paired columns compare matched straights in 2025 and 2026. "Earlier, harder / clipped": straights with clipping on which 2026 cars peak earlier and lose more speed, out of all straights with clipping. "Median diff. loss": median 2026 − 2025 difference in speed lost after the peak.

		2026 season			2025 vs 2026, paired		
Throttle (%)	Min. length (m)	Driver-straights	Peak before end (%)	Median peak position	Matched straights	Earlier, harder / clipped	Median diff. loss (km/h)
95	300	1320	70	0.88	61	45 / 46	+4.0
95	500	845	86	0.76	40	34 / 34	+8.0
95	800	395	95	0.66	18	18 / 18	+19.8
98	300	1316	70	0.88	60	44 / 44	+4.0
98	**500**	**842**	**86**	**0.76**	**39**	**33 / 33**	**+7.5**
98	800	395	95	0.66	18	18 / 18	+19.5
100	300	1245	69	0.88	59	44 / 44	+3.0
100	500	795	85	0.77	39	33 / 33	+7.0
100	800	370	94	0.66	18	18 / 18	+20.0

Note. The one straight with clipping that does not peak earlier and lose more speed in 2026 (95%, 300 m) is Silverstone T16: median peak position 0.97 in 2025 and 1.00 in 2026, median `clip_delta` 0.5 and 0.0 km/h.

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